EXP-4

Imputation techniques

Name-Shafiya Bennet Singh
Reg no-25BAI0199

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import numpy as np
import pandas as pd
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer, SimpleImputer

# Create sample DataFrame with missing values
df = pd.DataFrame(
    {
        "Name": ["Alice", "Bob", "Charlie", "David", "Emma"],
        "Age": [21, np.nan, 20, 23, np.nan],
        "City": ["Chennai", "Mumbai", np.nan, "Delhi", "Mumbai"],
    }
)

# 1. Mean Imputation (Numerical)
mean_imputer = SimpleImputer(strategy="mean")
df["Age_Mean"] = mean_imputer.fit_transform(df[["Age"]])

# 2. Median Imputation (Numerical)
median_imputer = SimpleImputer(strategy="median")
df["Age_Median"] = median_imputer.fit_transform(df[["Age"]])

# 3. Mode Imputation (Categorical)
mode_imputer = SimpleImputer(strategy="most_frequent")
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df["City_Mode"] = mode_imputer.fit_transform(df[["City"]]).ravel()

# 4. MICE Imputation (Numerical)
num_cols = ["Age"]
mice_imputer = IterativeImputer(max_iter=10, random_state=42)

df_mice = df[num_cols].copy()
df_mice.iloc[:, :] = mice_imputer.fit_transform(df_mice)

df["Age_MICE"] = df_mice["Age"]

print(df)
```